

Lampros Konstantellos

Electrical & Computer Engineer

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PROFILE

Electrical and Computer Engineer with research and consulting experience spanning PV, wind, BESS, EV charging and Vehicle-to-Grid (V2G) systems. Currently a Research Associate at the Center for Combined Smart Energy Systems (CoSES) and Doctoral Candidate (Dr.-Ing.) at the Chair of Renewable and Sustainable Energy Systems of the Technical University of Munich, researching Home Energy Management Systems in sector-coupled residential buildings. Experienced in real-time Hardware-in-the-Loop simulation, energy system modelling and optimization, techno-economic analysis and ML-based forecasting, as well as in technical due diligence and construction supervision of renewable energy projects. Author of three peer-reviewed publications, one of which received the 3rd Best Paper Award at IEEE PESS 2025.

PROFESSIONAL EXPERIENCE

Sep 2026 – Present

Research Associate

Technical University of Munich, Munich, Germany

Munich Institute of Integrated Materials, Energy and Process Engineering (MEP), Center for Combined Smart Energy Systems (CoSES)

- Conduct doctoral research on Home Energy Management Systems (HEMS), focusing on the coordination and optimization of distributed energy resources in sector-coupled residential buildings.
- Develop and evaluate energy management and optimization methods for smart energy systems and real-world applications.

May 2025 – Aug 2026

Renewable Energy Consultant

LDK Consultants S.A., Kifisia (Athens), Greece

- Performed technical due diligence and feasibility studies for PV, wind and BESS projects, including CAPEX/OPEX modelling and NPV, IRR, BCR, cash-flow and payback analysis.
- Conducted energy yield assessments (PVsyst) and reviewed civil, electrical and equipment designs to ensure project viability.
- Provided construction supervision and quality inspection services for wind farm projects (23.1 MW and 22.5 MW), including Factory Acceptance Tests, on-site testing review and commissioning oversight.
- Oversaw project preparation, procurement, EPC and O&M contract review, permitting, interconnection, scheduling and life-cycle assessment.
- Analysed business models and operational strategies to support investment decision-making.
- Prepared technical and financial proposals for international tenders, including IFI-financed (EBRD, EIB, World Bank) and private-sector projects.

Oct 2024 – Jan 2025

Project Intern – EV Charging Systems Analysis

Fraunhofer Institute for Solar Energy Systems ISE, Freiburg, Germany

Smart Grid Integration (SGI) team

- Evaluated commercial EV chargers under solar-optimized charging scenarios using Power Hardware-in-the-Loop testing in the Digital Grid Lab, within the BMWK-funded “Wallbox-Inspektion” project (FKZ 01MV23027).
- Developed and validated a Bidirectional GRU (BiGRU) neural network in Python forecasting five-hour charging behaviour from five-minute test data, achieving <5% energy prediction error.
- Quantified the financial impact of charging behaviour using 2023 German electricity prices and calculated CO₂ emissions based on national grid factors (0.354 kg/kWh).
- Research findings led to a publication at EVS38, the 38th International Electric Vehicle Symposium (Gothenburg, 2025).

Mar 2024 – Feb 2025

Master's Thesis Researcher

Technical University of Munich, Munich, Germany

Chair of Renewable and Sustainable Energy Systems (Prof. Thomas Hamacher), in formal co-supervision with the University of Patras (Prof. George Konstantopoulos)

- Built a real-time Hardware-in-the-Loop model (Typhoon HIL) of the 15 kV autonomous island grid of Kastellorizo, based on actual DSO topology and load measurements.
- Integrated a 0.5 MW PV plant and two 250 kW bidirectional EV chargers with an SOC-based V2G control algorithm, improving the grid frequency nadir by 20.6–21.9% under PV disconnection and sudden load disturbances.
- Reduced diesel generation by up to 33% and cut CO₂ emissions by ~360 kg per 3-hour cycle through distributed V2G active power injection (up to 80 kW support).
- Research findings led to a publication at IEEE PESS 2025 (**3rd Best Paper Award**).

Oct 2023 – May 2024

Laboratory Teaching Assistant

University of Patras, Patras, Greece

Department of Electrical and Computer Engineering

- Led laboratory exercises for the courses Electrical Power System Analysis and Power Systems Control & Stability, guiding groups of six students.
- Simplified complex concepts such as phase sequences, power flow and machine operation, earning positive feedback for clarity and hands-on teaching methods.

Jul 2023 – Sep 2023

Research Intern – Energy System Optimization

Technical University of Munich, Munich, Germany

Chair of Renewable and Sustainable Energy Systems (Prof. Thomas Hamacher)

- Developed cost-optimal energy transition scenarios for a medium-voltage island grid using the urbs linear programming model and the pyGRETA renewable assessment tool.
- Modelled the integration of 6.72 MW utility-scale PV, 2.41 MW rooftop PV, 1.12 MW wind and 51 MWh battery storage by 2052, achieving full decarbonization of the system and reducing electricity production costs by approximately €500/MWh under the optimal scenario.
- Contributed to the long-term grid planning strategy for non-interconnected Mediterranean island systems.

Sep 2022 – Dec 2025

Research Assistant (part-time)

University of Patras, Patras, Greece

Research group of Prof. Ioannis Tomkos

- Supported the implementation of EU-funded R&D projects (three EU Horizon projects), ensuring milestone compliance and reporting accuracy.
- Led communication, dissemination and exploitation activities, coordinating academic, industrial and municipal stakeholders.

Jun 2022 – Aug 2024

Research Assistant (part-time)

University of Patras, Patras, Greece

AI Hub (Prof. Epaminondas Mitronikas, Prof. Kyriakos Sgarbas)

- Contributed to the development of IoT-based real-time monitoring systems, integrating sensor hardware and communication protocols (Arduino, Modbus, MQTT).
- Supported the pilot deployment and validation of an intelligent water quality monitoring system for the Municipality of Western Achaia.

EDUCATION

Sep 2026 – Present

Doktor der Ingenieurwissenschaften (Dr.-Ing.)

Technical University of Munich, Munich, Germany

Doctoral Candidate, TUM School of Engineering and Design, Department of Energy & Process Engineering, Chair of Renewable and Sustainable Energy Systems

- Dissertation topic: *Home Energy Management Systems in Sector-Coupled Residential Buildings*.
- Supervised by Prof. Dr. rer. nat. Thomas Hamacher and mentored by Dr.-Ing. Anurag Mohapatra.

Diploma in Electrical and Computer Engineering

University of Patras, Patras, Greece

Integrated Master's degree, 5-year programme, 300 ECTS, EQF Level 7

- Final grade 7.54/10 (Very Good), thesis grade 10/10, ranked 7th in the 2019–2020 cohort.
- Diploma thesis: *Integration of Photovoltaic Power and Vehicle-to-Grid Technology in Electric Power Systems of Non-Interconnected Islands: A Case Study of Kastellorizo*.
- Supervised by Prof. George Konstantopoulos (University of Patras) and co-supervised by Prof. Thomas Hamacher (TUM).

PEER-REVIEWED PUBLICATIONS

- 2027 **L. Konstantellos**, A. Vazakas, K. Koumkoudis, N.-A. Papadopoulos, "Regulation-Aware BESS Dispatch for PV Systems under Virtual Net-Billing in Greece," *Electric Power Systems Research*, vol. 265, Part A, Art. no. 113914, 2027. [doi:10.1016/j.epsr.2026.113914](https://doi.org/10.1016/j.epsr.2026.113914)
- 2025 **L. Konstantellos**, A. Mohapatra, T. Kavvathas, G. Konstantopoulos, T. Hamacher, "Integration of PV and V2G Technology in an Island Grid: A Real-Time Simulation Study of Kastellorizo," *2025 IEEE Power and Energy Student Summit (PESS)*, Munich, Germany, pp. 31–36, 2025 (**3rd Best Paper Award**). [doi:10.30420/566656006](https://doi.org/10.30420/566656006)
- 2025 **L. Konstantellos**, Z. Kamacı, D. Pekmezci, B. Köpfer, "Financial Impact Analysis of Electric Vehicle Charging Behavior with RNN Model and Validation Against Real-World Data," *38th International Electric Vehicle Symposium and Exhibition (EVS38)*, Gothenburg, Sweden, 2025.

THESES & REPORTS

- 2025 **L. Konstantellos**, "Integration of Photovoltaic Power and Vehicle-to-Grid Technology in Electric Power Systems of Non-Interconnected Islands: A Case Study of Kastellorizo," *Diploma (Master's) Thesis*, University of Patras, 2025. [doi:10.5281/zenodo.14871102](https://doi.org/10.5281/zenodo.14871102)
- 2023 **L. Konstantellos**, "Novel Optimization Model Applied for Decarbonization Scenarios of Non-Interconnected Mediterranean Islands: A Kastellorizo Case Study," *Research Internship Report*, Technical University of Munich, 2023. [doi:10.5281/zenodo.13936256](https://doi.org/10.5281/zenodo.13936256)

TECHNICAL SKILLS

Programming & ML: Python (Pandas, NumPy, TensorFlow/Keras, Scikit-learn, Matplotlib), MATLAB, Simulink

Energy & Simulation Tools: PVsyst, Typhoon HIL, Power Hardware-in-the-Loop (PHIL), AutoCAD, QGIS, urbs, pyGRETA

Hardware & Protocols: Arduino, Modbus, MQTT

Software: LaTeX, MS Office, VS Code, PyCharm, Anaconda, Jupyter Notebook, Git

Collaboration Tools: Jira, Trello, Slack, MS Teams

LANGUAGES

Greek Native

English C2, Certificate of Proficiency in English (ECPE), University of Michigan (2017)

German B2, Goethe-Zertifikat B2, Goethe-Institut (2026)

PROFESSIONAL MEMBERSHIP

Member of the Technical Chamber of Greece (TEE), Reg. No. 171158